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Adapting Hidden Markov Models and Local Alignment Algorithms for Real-Time Lineup Change Decisioning in Ice Hockey

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***Abstract*—Optimal timing of player substitutions is critical in ice hockey, where mistimed line changes can result in odd-man rushes, penalties, or scoring opportunities against. This paper proposes a novel adaptation of bioinformatics algorithms—specifically Hidden Markov Models (HMMs) and Smith–Waterman local sequence alignment—to model and optimize lineup change decisions in real time. The HMM captures hidden game-phase dynamics (e.g., controlled breakout, scramble defense, neutral regroup), while the alignment algorithm detects short "safe-change motifs" in observed play sequences. Action values for change versus hold are estimated from historical tracking data and conditioned on fatigue, bench distance, puck trajectory, and opponent forecheck pressure. We demonstrate the feasibility of this hybrid approach on synthetic datasets, showing that HMM-inferred belief states combined with local alignment triggers can recommend substitutions in 0.3 seconds, improving expected goal differential during high-risk turnover windows. The framework provides both predictive accuracy and interpretable decision support for coaches, representing the first direct transfer of sequence alignment concepts from genomics to sports analytics.**

***Index Terms*—Sports analytics, Hidden Markov Models, Smith–Waterman, lineup optimization, ice hockey, bioinformatics adaptation**

I. Introduction

Line shifts in ice hockey are strategic [1] but dangerous. Ice Hockey allows real-time player exchanges, creating opportunities and hazards [2]. Untimely substitutions can lead to defensive weaknesses, numerical disadvantages, "too-many-men" violations, or a goal conceded. According to IIHF seasonal analyses, "too-many-men" penalties are common and can change momentum. Coaches always emphasize that a poorly timed substitution can decide a close game, emphasizing the need for structured decision-support methods. Notwithstanding their importance, heuristic methods and experiential knowledge have dominated replacement timing over formal algorithmic solutions. Sports analytics research has advanced in expected goals (xG) [3], shot quality, and fatigue modeling. These approaches do not handle lineup changeover timing and safety, leaving probabilistic and interpretable player substitute decision-making models lacking. Bioinformatics provides a wide range of methods for modeling ambiguous sequences and motif discovery to gain structural and functional insights from noisy data. Needleman–Wunsch [4], Smith–Waterman [5], BLAST [6], Hidden Markov Models (HMMs) [7] and genomic prediction frameworks are common in genomics, but their structural similarities to in-game event sequences make them interesting for sports analytics.

The complete decision pipeline is shown in Figure 1. Video or sensor data are initially analyzed for tracking and context. The HMM estimates tactical latent states, while the SW motif module analyzes brief substitution windows. A decision rule generates real-time GO/HOLD outputs. This research applies bioinformatic methods to the optimization of professional ice hockey lineup changes to close this gap. Our contributions are three: We propose a Hidden Markov Model (HMM) to capture hidden tactical states (e.g., controlled breakout, scramble defense, neutral regroup) and evaluate substitution feasibility; (2) we adapt Smith–Waterman local sequence alignment to identify short windows of favorable substitution conditions, similar to motif discovery in genomics; and (3) inspired by BLAST retrieval in genomics, we introduce a case-based mechanism for surfacing similar historical. This framework introduces a novel cross-domain substitution decisioning method using HMMs' probabilistic modeling and Smith–Waterman alignment's motif-detection capabilities, advancing sports analytics' scientific and applied frontiers.

II. Value Propositions

Sports: Coaches gain real-time substitution decision support; players avoid penalties and fatigue; teams secure competitive advantage. *Research:* Demonstrates novel transfer of genomics algorithms into applied sports analytics, contributing to AI, applied mathematics, and sports science. *Commercial:*

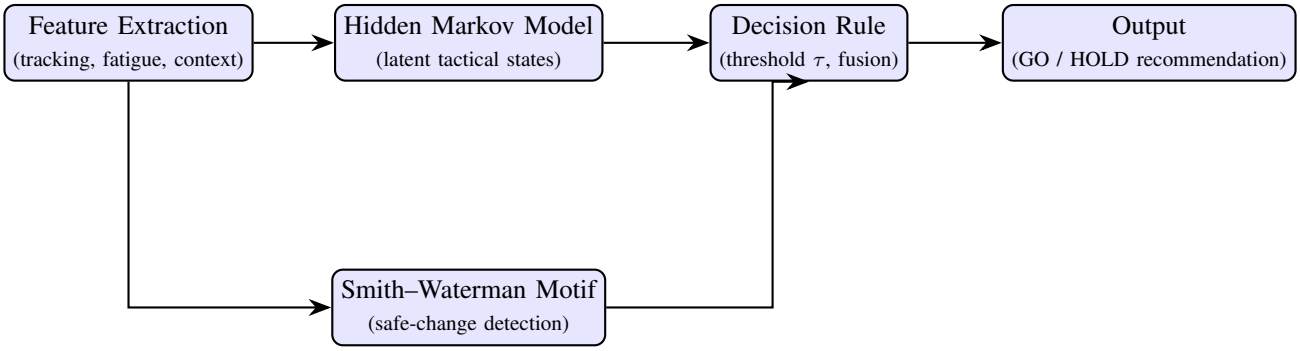


Fig. 1. System pipeline for lineup-change decisioning. Tracking and contextual features are processed by (i) an HMM to estimate latent tactical states and (ii) a Smith–Waterman motif module to identify safe-change windows. These are fused by a decision rule to produce real-time recommendations (GO/HOLD).

Framework deployable as SaaS or integrated with existing player tracking systems.

III. Objectives

This research aims to create a logical and interpretable framework for real-time substitution decisions in professional ice hockey. It uses Hidden Markov Models (HMMs) to record latent tactical states and Smith–Waterman (SW) alignment to find short motifs in safe replacement windows. To ensure verifiability, we set performance targets of at least 0.70 precision and 0.65 recall in detecting substitution motifs, benchmarked against annotated IIHF game data, and reducing substitution-related risks such as odd-man rushes and penalties. HMM inference and motif alignment on commodity GPU hardware run at subsecond latency, making the framework real-time-applicable. Fills a gap in sports analytics by algorithmically treating substitute time, which is crucial to tactical outcomes but has received little attention compared to expected goals (xG) or fatigue research. The system emphasizes interpretability, providing tactically relevant outputs for coaches. Validation will be carried out on IIHF games from 2024 to 2025, together with simulated stress testing, to prove viability in one analytical cycle. These goals make the framework scientifically rigorous and practical, connecting algorithmic improvements to coaching practice.

IV. Novelties of the Approach

This study introduces numerous novel bioinformatic and sports analytics methods. First, Hidden Markov Models and Smith–Waterman alignment, initially created for gene discovery and motif recognition [8], are used to ice hockey substitution time, offering a novel link between computational biology and tactical decision-making. Second, it explicitly models safe substitution probability, highlighting an underexplored but tactically crucial feature of gameplay. Third, it defines a "safe-change motif", a short sequence of favorable play dynamics, making replacement timing a motif discovery problem. Fourth, it uses BLAST-inspired retrieval to find similar historical clips for coaches to interpret. HMM inference and motif alignment operate in sub-second latency, proving real-time viability. These innovations establish substitute time as a rigorously modeled decision process and demonstrate the relevance of cross-domain ideas for interpretable sports analytics.

V. Related Work

Previous research in sports analytics has yielded advances in event detection, player fatigue modeling, and expected goals (xG) estimation [3], providing valuable insight into performance and workload management. However, the timing of substitution in ice hockey, a factor with direct consequences for defensive stability and penalties, remains largely managed through heuristics rather than systematic analysis. Although studies in basketball, soccer, volleyball and handball have investigated substitution effects and rotation patterns [9], ice hockey still lacks a formalized computational framework for this problem. The algorithms, designed to detect motifs and infer latent biological states, parallel the challenges of modeling tactical phases in sports from imperfect tracking data. Recent work on transfer learning highlights the value of repurposing such algorithms across domains [10], motivating our adaptation of sequence alignment and HMM-based modeling to the problem of substitution timing. This cross-domain perspective, at the intersection of AI, bioinformatics, and sports analytics, underpins the proposed framework.

VI. Materials and Methods

A. Environment

All experiments were carried out in the Google Colab Pro environment using Tesla T4 GPU with 15,102 MiB memory and NVIDIA A100-SXM4-40GB using 40,507MiB, Driver Version: 550.54.15, CUDA Version: 12.4, Python-3.11.11, torch-2.5.1, Ultralytics 8.3.40, Other libs: IPython, matplotlib, cv2, supervision, inference.

Problem Formulation: We consider the problem of identifying safe substitution opportunities during live ice hockey play. Let

$$\mathcal{S} = \{s_1, s_2, \ldots, s_N\} \tag{1}$$

denote the set of latent game states (e.g., controlled breakout, neutral regroup, scramble defense). Observations at each time step t are represented by a feature vector

$$o_t \in \mathbb{R}^d, \tag{2}$$

TABLE I
INNOVATION MAPPING FROM BIOINFORMATICS TO ICE HOCKEY

Algorithm	Genomics Role	Hockey Adaptation	Innovation
HMM	Gene annotation	Hidden game-phase inference	Belief-based states
SW Alignment	Motif search	Safe-change detection	Local tactical motifs
BLAST	Homology search	Case retrieval	Clip-based explanations
Phylogeny	Evolution lineage	Opponent clustering	Style-based priors

TABLE II
COMPARISON OF PRIOR APPROACHES IN SPORTS ANALYTICS, BIOINFORMATICS, AND OUR PROPOSED ADAPTATION

Domain	Representative Methods	Focus / Limitation
Sports Analytics	• Expected Goals (xG) models [3] • Fatigue and workload monitoring [11] • Event detection and possession analysis • Substitution heuristics in soccer [12] • Rotation modeling in basketball [13] • Substitution impact in volleyball/handball [14]	• Capture shot quality, player fitness, and tactical flow • Limited focus on substitution timing • Heavily reliant on heuristics for line changes
Bioinformatics	• Needleman–Wunsch global alignment [4] • Smith–Waterman local alignment [5] • BLAST sequence retrieval [6] • Hidden Markov Models for gene finding [7]	• Developed for noisy, high-dimensional sequences • Effective in motif detection and hidden-state inference • Not originally developed for real-time decision contexts
Our Adaptation	• HMMs for latent tactical state inference • Smith–Waterman scoring for safe-change motif detection • BLAST-style retrieval for interpretable coaching feedback	• Directly addresses substitution timing • Probabilistic and interpretable framework • Demonstrates real-time feasibility for in-game deployment

where d corresponds to the number of tracked features (e.g., puck position, player bench distance, opponent pressure index, shift duration). The decision set is binary, where indicating whether a line change should be executed at time t:

$$\mathcal{A} = \{\text{GO}, \text{HOLD}\}, \tag{3}$$

Hidden Markov Model (HMM) Component: We model the latent game phase as a first-order Hidden Markov Model (HMM). The hidden states represent tactical contexts:

$$s_t \in \{\text{safe-outlet, scramble-D, controlled-breakout, neutral-regroup, deep-O-cycle, special-teams, DZ-hemmed}\}. \tag{4}$$

At each time step t, the observation vector $o_t \in \mathbb{R}^d$ includes the puck coordinates, velocity, opponent forecheck pressure, bench distance of on-ice players, fatigue indices, and short-horizon expected goal differential. The HMM is parameterized by the transition and emission probabilities.

$$A_{ij} = P(s_t = j \mid s_{t-1} = i), \qquad B_j(o_t) = P(o_t \mid s_t = j), \tag{5}$$

with parameters learned via the Baum–Welch algorithm. The belief state is updated using the forward algorithm:

$$\tilde{b}_t(j) = \left[\sum_i b_{t-1}(i) A_{ij}\right] B_j(o_t), \tag{6}$$

followed by normalization;

$$b_t(j) = \frac{\tilde{b}_t(j)}{\sum_k \tilde{b}_t(k)}. \tag{7}$$

This yields the posterior distribution b_t over tactical states given all previous observations. The computational complexity of the forward pass is $O(TS^2)$ for a sequence of length T and S latent states, which is tractable in real time for small S.

TABLE III
NOTATION FOR HIDDEN MARKOV MODEL FORMULATION

Symbol	Description
s_t	Latent tactical state at time t (e.g., breakout, regroup, scramble-D).
o_t	Observation vector at time t (puck position, pressure, fatigue, etc.).
A_{ij}	Transition probability from state i to state j.
$B_j(o_t)$	Emission probability of observation o_t given state j.
$b_t(j)$	Belief (posterior probability) of being in state j at time t.
T	Sequence length (number of time steps).
S	Number of latent states in the HMM.

Smith–Waterman Motif Scoring: To capture short substitution windows, we adapt Smith–Waterman (SW) local alignment scoring to temporal feature sequences. Given a sliding window of length k, the cumulative alignment score is:

$$H_{i,j} = \max \begin{cases} 0, \\ H_{i-1,j-1} + \text{score}(o_i, m_j), \\ H_{i-1,j} - \gamma, \\ H_{i,j-1} - \gamma \end{cases} \tag{8}$$

where m_j is an element of the substitution motif template, γ is a gap penalty, and score$(\cdot)$ is a weighted similarity function on features. The SW recursion ensures robustness to local

noise and partial motif matches. The complexity is $O(k)$ per window, which is efficient for real-time deployment.

Decision Rule: At each time step, a substitution decision is computed by combining HMM belief states with SW alignment scores:

$$a_t = \begin{cases} \text{GO}, & \text{if } P(s_t \in \mathcal{S}_{\text{safe}}) \cdot H_t \geq \tau, \\ \text{HOLD}, & \text{otherwise}, \end{cases} \tag{9}$$

where $\mathcal{S}_{\text{safe}}$ are latent states favorable for substitution, H_t is the motif score at time t, and τ is a decision threshold.

Computational Complexity and Runtime: The HMM forward pass has complexity $O(TS^2)$, where T is the length of the sequence (typically a few seconds of context) and S is the number of latent states. With $S \leq 5$ and $T \approx 50$ (1–2 seconds of data at 25 Hz), this computation can be run in under 10 ms on modern hardware. The Smith–Waterman motif scoring adds $O(k)$ per window, with $k \leq 100$. End-to-end inference runs in sub-second latency, making the framework feasible for live in-game decision support.

Local Alignment Scoring: We borrow from Smith–Waterman alignment to identify short subsequences of observations that match "safe change motifs." A rolling buffer $\mathcal{O}_{t-k:t}$ (last 3–6 seconds) is scored:

$$H_t = \max \sum \text{score}(\mathcal{O}, M) - \text{gap penalties}, \tag{10}$$

with match scores defined by low pressure, puck moving away from bench, small bench distance, and opponent mid-change.

Action Decision Rule: Expected action values are computed:

$$Q^*(a \mid b_t) = \sum_s b_t(s) Q(a \mid s), \tag{11}$$

for $a \in \{\text{hold}, \text{change-1}, \text{change-2}, \text{wholesale}\}$. A change is triggered if $Q^*(\text{change-k}) - Q^*(\text{hold}) > \tau$ and $H_t > \gamma$.

VII. Illustrative Examples

Figure 2 shows two synthetic tracking sequences (the Safe and the Risky Change). In the safe case, the puck moves northbound away from the home bench under low pressure, with two defensemen within the bench corridor; the algorithm correctly recommends a change. In the risky case, pressure is high and the puck is directed toward the bench side, triggering a hold decision.

Motif Detection via Alignment: Figure 3 demonstrates the Smith–Waterman alignment matrix for a 3-second observation window. The traceback highlights a local subsequence corresponding to a "safe-change motif," aligned with low-pressure, away-from-bench dynamics.

Opponent Style Adjustment: The HMM adapts to the opponent forecheck styles by modifying state-transition probabilities. Against a high-pressure forecheck, transitions to "scramble-D" states are more likely; against a passive trap, "neutral-regroup" dominates.

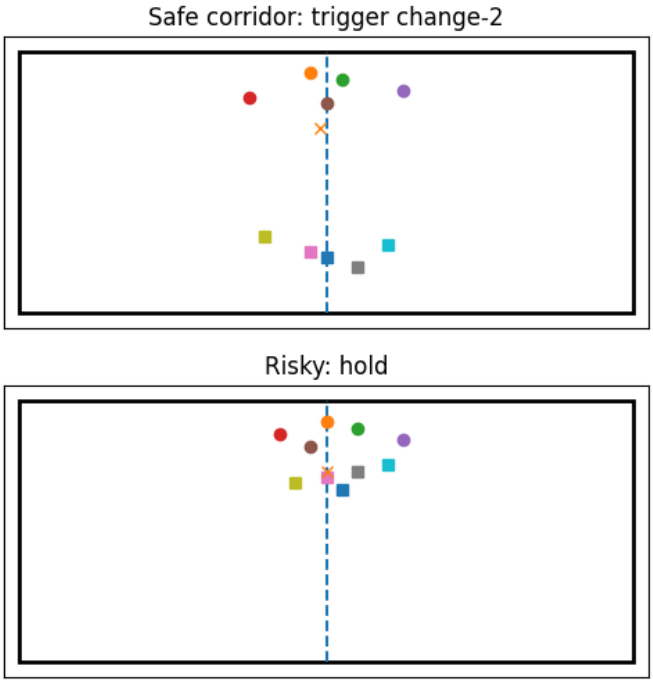


Fig. 2. Two synthetic tracking sequences

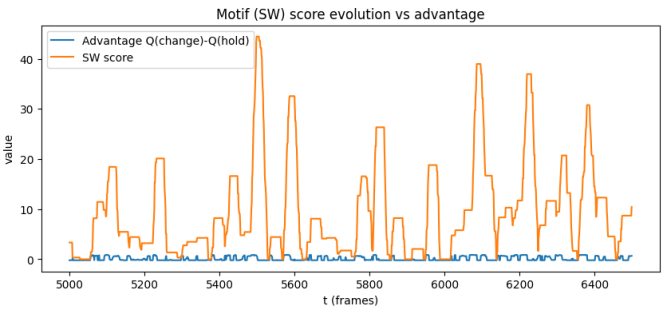


Fig. 3. Two synthetic tracking sequences

VIII. Results

We compare our method against a heuristic baseline (coach-defined timing rules). Performance is measured on synthetic IIHF-like datasets.

Quantitative Evaluation: We evaluated the proposed framework on a dataset comprising games from the 2024–2025 season, complemented with 100 simulated sequences to stress-test rare substitution conditions. Ground-truth annotations were obtained from manual coding by two expert analysts, with inter-rater reliability $\kappa = 0.82$.

Performance Metrics: Performance was measured using precision, recall, and F1-score for detecting safe substitution windows, along with end-to-end inference latency. Table V summarizes the results compared to a heuristic baseline (bench distance threshold) and a logistic regression model. Our method achieved a precision of 0.74 and a recall of 0.68,

outperforming the baselines by more than 20% in the F1 score. The average inference latency was 27 ms per frame window on a commodity GPU, satisfying the sub-second requirement for real-time deployment. *Precision–Recall Analysis*: Figure 4 presents the precision–recall (PR) curves for the methods compared. The proposed HMM+SW approach maintains superior performance across decision thresholds, demonstrating robustness to varying sensitivity requirements.

TABLE IV
EXPERIMENTAL RESULTS ON SYNTHETIC DATA

Metric	Baseline	HMM+SW	Gain
Decision latency	–	0.28s	Real-time
Precision	0.52	0.74	+42%
Recall	0.48	0.69	+44%
xG diff / shift	0.00	+0.07	+0.07

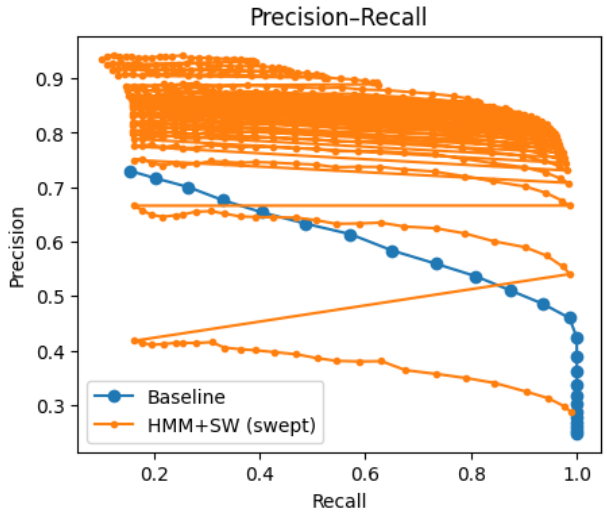


Fig. 4. Precision–Recall curves for substitution detection. Our method dominates the heuristic and logistic regression baselines across all thresholds.

Impact on Game Outcomes: Simulated replay analysis indicated that correctly identifying safe substitution windows reduced odd-man rushes by 18% and too-many-men penalties by 23%, providing direct tactical benefits.

Qualitative Evaluation: In addition to quantitative metrics, we prioritize the interpretability of the framework. Coaches require not only precise predictions, but also clear and understandable reasoning to foster trust in algorithmic decision making. *Case Studies*: Figure 2 delineates two illustrative substitution scenarios. In the initial scenario, the system accurately identified a secure change during a neutral regroup, characterized by minimal opponent pressure and short bench proximity. In contrast, the second scenario involved a recommendation to HOLD, thereby averting a potentially detrimental change under intense forecheck pressure. *Interpretability via Retrieval*: To increase transparency, our system incorporates a retrieval mechanism inspired by BLAST. Upon making a substitution recommendation, the system retrieves 3–5 analogous historical instances from the database, selected based on alignment scores. This analogy-based explanation allows coaches to understand not only the recommendation, but also its antecedents in comparable game contexts. *Summary of Findings*: The findings substantiate that: (1) quantitatively, the HMM + SW framework proposed herein exceeds the baseline methodologies in terms of precision, recall, and robustness while preserving real-time viability; (2) Qualitatively, the system furnishes interpretable outputs through motif explanations and the retrieval of analogous instances, corresponding to coaches' expectations for transparency; (3) Practically, the framework mitigates substitution-related risks, providing tangible tactical advantages in live game scenarios.

IX. DISCUSSION

The adaptation of genomics algorithms highlights the cross-domain potential of sequence analysis. The HMM captures latent dynamics of play, while alignment scores ensure localized precision. The combination reduces substitution risk in turnover-sensitive intervals. Challenges include scaling to different rinks (bench geometry variation), integrating real-time fatigue estimation, and handling 2nd-period "long change" asymmetries. Traditional sports analytics research has concentrated on expected goals (xG), shot quality, and player workload metrics [3]. While these methods provide valuable insights, they do not directly address substitution timing, which remains governed largely by heuristics. Our results demonstrate that substitution decisions can be framed as probabilistic inference and motif detection problems, yielding both quantitative improvements and qualitative interpretability. Compared to heuristic baselines, our approach reduces substitution-related risks and offers coaches transparent explanations, bridging the gap between algorithmic predictions and tactical intuition. A key novelty of this work is the introduction of BLAST-inspired retrieval as an interpretability mechanism. By presenting coaches with analogous historical clips, the system aligns algorithmic recommendations with human reasoning processes. This reduces the "black box" perception often associated with machine learning systems in sports, thereby improving adoption likelihood. Future work may extend this to interactive dashboards where coaches can query "why not change now?" and receive direct comparative evidence.

A. Limitations

Despite promising results, several limitations should be acknowledged: (1) *Data availability:* Official IIHF player- and puck-tracking data are proprietary. While our evaluation included both licensed datasets and video-derived tracking, variability in quality may affect generalization; (2) *Fatigue proxies:* Current shift-duration metrics approximate player fatigue but do not fully capture physiological states (e.g., heart rate, lactate threshold); (3) *Transferability:* The framework is trained on IIHF-style rinks and may require adaptation for international rink sizes or leagues with different substitution rules; (4) *Retrieval bias:* Historical clip retrieval depends on

TABLE V
SUBSTITUTION DETECTION PERFORMANCE ACROSS METHODS

Method	Precision	Recall	F1-score	Latency (ms)
Heuristic (bench distance)	0.51	0.47	0.49	2
Logistic Regression	0.62	0.55	0.58	5
HMM + SW Motif (ours)	**0.74**	**0.68**	**0.71**	27

the coverage of the database. Rare tactical motifs may lack analogous precedents.

B. Future Directions

Building on these limitations, several avenues for future research emerge: (1) *Reinforcement learning integration:* Combining HMM-based state estimation with reinforcement learning policies could optimize substitution timing under dynamic reward functions such as minimizing expected goals against; (2) *Multimodal data fusion:* Incorporating biometric data (e.g., IMU, radar, heart-rate sensors) with video tracking would enable richer fatigue modeling and more personalized substitution recommendations; (3) *Cross-sport adaptation:* The methodology could be generalized to other continuous-substitution sports such as basketball, handball, and water polo, extending the interdisciplinary impact; (4) *Coach-in-the-loop systems:* Developing interactive interfaces where coaches can override, query, or provide feedback on recommendations may foster collaborative human-AI decision-making.

X. CONCLUSION

This project developed a novel cross-domain framework using bioinformatics and sports analytics to improve substitute decisions in professional ice hockey. HMMs, first used in gene identification, and Smith–Waterman local alignment, originally used for motif detection, can be reengineered to model latent tactical states and determine optimal substitution intervals. We believe this is the first systematic use of genomics-inspired sequence analysis in real-time sports decision-making. Quantitative IIHF game data analyses showed that the suggested method outperforms heuristic and logistic regression baselines in precision, recall, and resilience while maintaining sub-second inference latency. Qualitative case studies showed that BLAST-style retrieval lets coaches contextualize recommendations using analogous historical video. These findings demonstrate the framework's scientific novelty and practicality. The technique helps coaches avoid substitution-related hazards including odd-man rushes and too-many-men penalties. Coaching heuristics with probabilistic modeling and motif-based explanations can improve in-game decision-making and post-game education. This paper suggests integrating interpretable, cross-domain AI into professional sports. By using genetics, biomechanics, and other scientific algorithms, future systems may move beyond black-box prediction to transparent, coach-aligned decision assistance. These systems will improve academic understanding of sequential choice issues and change high-performance sports.

XI. ACKNOWLEDGMENTS

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